Partial Derivatives and the Gradient

ANSWER KEY

First and second partial derivatives

1 For $f(x, y) = x^3y^2 - 4xy + y$, find:

a $f_x = 3x^2y^2 - 4y$

b $f_y = 2x^3y - 4x + 1$

c $f_{xx} = 6xy^2$

d $f_{xy} = 6x^2y - 4$

2 For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

$f_x = ye^{xy}\sin(x + y) + e^{xy}\cos(x + y)$.

The gradient and directional derivatives

3 Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

$\nabla f = \langle 2xy, x^2 + 9y^2\rangle$. At $(1, 2)$: $\nabla f(1, 2) = \langle 4, 37\rangle$.

4 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4\rangle$.

$\nabla f = \langle 2x, 2y\rangle$, so $\nabla f(1, 1) = \langle 2, 2\rangle$. Unit vector: $\hat{\mathbf{v}} = \langle \frac{3}{5}, \frac{4}{5}\rangle$. $D_{\mathbf{v}}f = \langle 2, 2\rangle \cdot \langle \frac{3}{5}, \frac{4}{5}\rangle = \frac{6}{5} + \frac{8}{5} = \frac{14}{5}$.

Tangent planes

5 Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

$f_x = 2x$, $f_y = 2y$: at $(1, 2)$, $f_x = 2$, $f_y = 4$. Tangent plane: $z - 5 = 2(x - 1) + 4(y - 2)$, i.e. $z = 2x + 4y - 5$.

6 Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

$f_x = y$, $f_y = x$: at $(2, 3)$, $f_x = 3$, $f_y = 2$. Tangent plane: $z - 6 = 3(x - 2) + 2(y - 3)$, i.e. $z = 3x + 2y - 6$.

Critical points of multivariable functions

7 Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.

$f_x = 2x - 4 = 0$ gives $x = 2$; $f_y = 2y + 2 = 0$ gives $y = -1$. Critical point: $(2, -1)$.

8 Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

$f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$: $D = 2(2) - 0 = 4 > 0$ and $f_{xx} = 2 > 0$, so $(2, -1)$ is a local minimum.

The chain rule for multivariable functions

9 Let $z = x^2 y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.
$\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. At $t = 1$: $x = 1, y = 1$. $\frac{dz}{dt} = 2(1)(1)(2) + 1(3) = 4 + 3 = 7$.

10 Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.
$\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = 2z$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$.
$\frac{dw}{dt} = 2x(-\sin t) + 2y\cos t + 2z = -2\cos t\sin t + 2\sin t\cos t + 2t = 2t$ (the first two terms cancel).

Maximum rate of increase

11 Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.

$\nabla f = \langle 2x + y, x\rangle$; at $(1, 2)$: $\nabla f = \langle 4, 1\rangle$. Direction of steepest increase: $\langle 4, 1\rangle$ (or its unit vector). Maximum rate of increase: $|\nabla f| = \sqrt{16 + 1} = \sqrt{17}$.

12 Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

$\nabla f = \langle 2x, 2y\rangle$; at $(3, 4)$: $\langle 6, 8\rangle$. Steepest descent is the opposite direction: $\langle -6, -8\rangle$ (pointing toward the origin, the minimum).

Implicit partial differentiation

13 For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).
$2x + 2z\frac{\partial z}{\partial x} = 0$, so $\frac{\partial z}{\partial x} = -\frac{x}{z}$.

Classifying critical points with saddle points

14 Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

$f_x = 2x = 0$, $f_y = -2y = 0$ gives $(0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$: $D = (2)(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point (neither a local max nor min).

15 Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

$f_x = 3x^2 - 3y = 0$ gives $y = x^2$. $f_y = -3x + 3y^2 = 0$ gives $x = y^2 = x^4$, so $x^4 - x = 0$, $x(x^3 - 1) = 0$: $x = 0$ or $x = 1$. Critical points $(0, 0)$ and $(1, 1)$. $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$. At $(0, 0)$: $D = 0 - 9 = -9 < 0$: saddle point. At $(1, 1)$: $D = (6)(6) - 9 = 27 > 0$ and $f_{xx} = 6 > 0$: local minimum.

An applied optimization problem

16 A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.

$P_x = 100 - 2x - y = 0$; $P_y = 120 - 4y - x = 0$. From the first: $y = 100 - 2x$. Substituting: $120 - 4(100 - 2x) - x = 0$: $120 - 400 + 8x - x = 0$: $7x = 280$, $x = 40$. Then $y = 100 - 80 = 20$. Maximum profit at $(x, y) = (40, 20)$.

Higher-order mixed partial derivatives

17 For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).

$f_x = 3x^2y^2 + y\cos(xy)$: $f_{xy} = 6x^2y + \cos(xy) - xy\sin(xy)$. $f_y = 2x^3y + x\cos(xy)$: $f_{yx} = 6x^2y + \cos(xy) - xy\sin(xy)$. Both equal, confirming Clairaut's theorem.

Linear approximation in two variables

18 Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

$f(2, 3) = 12$. $f_x = 2xy$: $f_x(2, 3) = 12$. $f_y = x^2$: $f_y(2, 3) = 4$. Estimate: $12 + 12(0.1) + 4(-0.1) = 12 + 1.2 - 0.4 = 12.8$.

19 The volume of a cylinder is $V(r, h) = \pi r^2h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\,dr + V_h\,dh$ to estimate the maximum possible error in the computed volume.

$V_r = 2\pi rh$: at $(3, 10)$, $V_r = 60\pi$. $V_h = \pi r^2$: at $(3, 10)$, $V_h = 9\pi$. Maximum error estimate: $|dV| \le |V_r||dr| + |V_h||dh| = 60\pi(0.05) + 9\pi(0.1) = 3\pi + 0.9\pi = 3.9\pi \approx 12.25$ cm^3.

20 For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$ $f_x = 2xy - 3y^2$: $f_x(1, -1) = 2(1)(-1) - 3(1) = -2 - 3 = -5$.

b $f_y(1, -1)$ $f_y = x^2 - 6xy + 3y^2$: $f_y(1, -1) = 1 - 6(1)(-1) + 3(1) = 1 + 6 + 3 = 10$.

c The gradient vector $\nabla f(1, -1)$ $\nabla f(1, -1) = \langle -5, 10\rangle$.

d The directional derivative in the direction of $\langle 1, 0\rangle$ $D_{\langle 1, 0\rangle}f = \nabla f \cdot \langle 1, 0\rangle = -5$ (since $\langle 1, 0\rangle$ is already a unit vector).